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Bellman error centering

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ABSTRACT

This paper re-examines recent reward centering methods in reinforcement learning (RL), focusing on Simple Reward Centering (SRC) and Value-based Reward Centering (VRC). Our analysis distinguishes SRC as direct reward centering, while proving VRC's mathematical equivalence to Bellman Error Centering (BEC)—a novel reinterpretation that clarifies its underlying mechanism. Leveraging BEC, we derive two theoretical advances: (1) a centered fixed-point solution for tabular value functions, and (2) a centered TD fixed-point solution under linear function approximation. We propose two algorithms to realize these: on-policy Centered Temporal Difference (CTD) and off-policy Centered Temporal Difference with Correction (CTDC), with convergence proofs under standard assumptions. Experiments on benchmark environments show our methods outperform baselines in stability and performance. The BEC paradigm facilitates seamless integration with existing RL architectures, enabling broad applicability.

1. Introduction

Reinforcement learning (RL) has achieved breakthroughs in strategic decision-making (e.g., AlphaGo [Chen & Yu, 2016](#); [Silver et al., 2016](#)) and human-aligned large language models (e.g., ChatGPT via RLHF [Carta et al., 2023](#); [Dai et al., 2024](#); [Guo et al., 2025](#); [Ouyang et al., 2022](#)). Yet, these advances demand immense computational resources: AlphaZero requires billions of environment interactions, and RL training of state-of-the-art models consumes millions of GPU hours ([Patterson et al., 2021](#)). This underscores the need for efficiency improvements in RL to support scalable AI.

Long-horizon and continuous RL problems are typically addressed via average-reward or discounted-reward criteria. For average rewards, methods like R-Learning ([Schwartz, 1993](#)), SMART ([Das et al., 1999](#)), RVI Q-Learning ([Abounadi et al., 2001](#)), CSV-Learning ([Yang et al., 2016](#)), and Differential Q/TD-Learning ([Wan et al., 2021](#)) estimate and subtract average rewards to stabilize learning. For discounted rewards, approaches such as reward shifting ([Sun et al., 2022](#)) and near-Blackwell-optimal adjustments ([Schneckenreither & Moser, 2025](#)) mitigate issues with large discount factors. Recently, [Naik \(2024\)](#), [Naik et al. \(2024\)](#) introduced Reward Centering, including Simple Reward Centering (SRC) for on-policy and Value-based Reward Centering (VRC) for off-policy settings, inspired by [Wan et al. \(2021\)](#). These subtract estimated average rewards to enhance performance in continuous tasks,

robustness to reward offsets, and convergence in tabular Q-Learning, DQN, and linear approximations.

Despite these advances, three challenges persist: (1) Integrating Reward Centering with other RL algorithms is non-trivial, as its mechanisms remain underexplored ([Naik, 2024](#); [Naik et al., 2024](#)); (2) Convergence in large state spaces with function approximation is unproven; (3) The converged solution is unclear.

To address these, our contributions are:

(1) We prove VRC is equivalent to Bellman Error Centering (BEC), distinguishing it from SRC and revealing its core as error rather than reward adjustment—unlike prior methods like Differential TD, which lack this reinterpretation.

(2) Under linear approximation, BEC converges to a centered TD fixed point, extending beyond tabular proofs in [Naik \(2024\)](#), [Naik et al. \(2024\)](#).

(3) We design on-policy CTD and off-policy CTDC algorithms, with convergence guarantees under standard assumptions.

(4) Experiments demonstrate superior stability and performance over baselines.

In the early modeling phase, we encountered challenges in proving the mathematical equivalence between BEC and VRC due to intricate derivations involving infinite-horizon MDPs. These were overcome through rigorous fixed-point analysis, ensuring theoretical soundness. For method design, extending to off-policy settings proved difficult, as

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standard TD methods risk divergence; we addressed this by incorporating correction terms inspired by Differential Q-learning (Wan et al., 2021). During experiments, we faced instabilities from divergent TD behaviors, a known issue in the literature (e.g., Baird's counterexample (Baird et al., 1995)). This was mitigated via our centered update rules and by providing reproducible code at [https://github.com/GameAI-NJUPT/BEC].

2. Background

2.1. Markov decision process

Reinforcement learning agent keeps interactions with an environment. In each interaction t , it observes the state s_t , takes an action a_t to influence the environment, and obtains an immediate reward r_{t+1} . Consider an infinite-horizon Markov Decision Process (MDP), defined by a tuple $\langle S, \mathcal{A}, \mathcal{R}, \mathcal{P} \rangle$, where $S = \{1, 2, \dots, N\}$ is a finite set of states of the environment; $\mathcal{A}$ is a finite set of actions of the agent; $\mathcal{R} : S \times \mathcal{A} \times S \rightarrow \mathbb{R}$ is a bounded deterministic reward function; $\mathcal{P} : S \times \mathcal{A} \times S \rightarrow [0, 1]$ is the transition probability distribution (Sutton & Barto, 2018).

A policy is a mapping $\pi : S \times \mathcal{A} \rightarrow [0, 1]$. The goal of the agent is to find an optimal policy π^* to maximize the expectation of discounted cumulative rewards over a long period. State value function $V^\pi(s)$ for a stationary policy π is defined as:

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid S_0 = s \right], \quad (1)$$

where $\gamma \in (0, 1)$ is a discount factor. We have the Bellman equation for each state:

$$V^\pi(s) = \mathbb{E}_\pi [r + \gamma V^\pi(s')], \quad (2)$$

where s' is the successor state of state s . In vector form,

$$\begin{aligned} \mathbf{V}^\pi &= \mathbf{R}^\pi + \gamma \mathbb{P}^\pi \mathbf{V}^\pi \\ &\doteq \mathcal{T}^\pi \mathbf{V}^\pi, \end{aligned} \quad (3)$$

where $R^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s, a, s') R(s, a, s')$, transition probability from state s to state s' is $[\mathbb{P}^\pi]_{s,s'} = \sum_a \pi(a|s) \sum_{s'} P(s, a, s')$ where $\mathbb{P}^\pi$ represents the transition probability matrix under policy π , and $\mathcal{T}^\pi$ is Bellman prediction operator. $\mathbf{V}^\pi$ is the fixed point solution of $\mathbf{V} = \mathcal{T}^\pi \mathbf{V}$. In the Markov decision process, the stationary distribution $\mathbf{d}_\pi$ refers to the probability distribution of states that remains unchanged over time under the policy π , satisfying $(\mathbb{P}^\pi)^\top \mathbf{d}_\pi = \mathbf{d}_\pi$.

To deal with large-scale state space, linear value function for state $s \in S$ is defined as:

$$V_\theta(s) := \boldsymbol{\theta}^\top \boldsymbol{\phi}(s) = \sum_{i=1}^m \theta_i \phi_i(s), \quad (4)$$

$V_\theta(s)$ is a parametric approximation of the true state-value function $V(s)$ where $\boldsymbol{\theta} := (\theta_1, \theta_2, \dots, \theta_m)^\top \in \mathbb{R}^m$ is a parameter vector, $\boldsymbol{\phi} := (\phi_1, \phi_2, \dots, \phi_m)^\top \in \mathbb{R}^m$ is a feature function defined on state space S , and $m \ll |S|$ is the feature size. In vector form, $\mathbf{V}_\theta = \mathbf{\Phi} \boldsymbol{\theta}$, where $\mathbf{\Phi} \in \mathbb{R}^{|S| \times m}$ denotes the feature matrix. It constitutes standard practice to augment the feature space with a constant basis function, specifically designed to account for the bias term in the parametric model, thereby enhancing the estimator's representational capacity.

However, in the parameter space, $\mathbf{V}_\theta$ cannot be guaranteed to be equal to $\mathcal{T}^\pi \mathbf{V}_\theta$. We typically solve for the TD fixed point (Chen et al., 2023b; Sutton et al., 2009, 2008), as follows:

$$\mathbf{V}_\theta = \Pi \mathcal{T}^\pi \mathbf{V}_\theta, \quad (5)$$

where projection $\Pi = \mathbf{\Phi}(\mathbf{\Phi}^\top \mathbf{D} \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{D}$, $\mathbf{D}$ is a diagonal matrix of distribution vector $\mathbf{d}$, each element d_s represents the distribution of state s . In expectation form, it is equal to

$$\mathbb{E}[\delta \boldsymbol{\phi}] = 0, \quad (6)$$

where TD error $\delta = r + \gamma V_\theta(s') - V_\theta(s)$.

In on-policy learning, the target policy π and the behavior policy μ are the same, and the experience is sampled as $\langle s_t, a_t, r_{t+1}, s_{t+1} \rangle$. The update rule is (Sutton et al., 2016)

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha_t \delta_t \boldsymbol{\phi}_t, \quad (7)$$

where $\alpha_t \in (0, 1)$ is a learning stepsize, and $\delta_t = r_{t+1} + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)$.

In off-policy learning, the target policy π and the behavior policy μ are different. The update rule of the off-policy TD learning with importance sampling ratio $\rho_t = \frac{\pi(a_t|s_t)}{\mu(a_t|s_t)}$ (Sutton et al., 2016) is

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha_t \rho_t \delta_t \boldsymbol{\phi}_t. \quad (8)$$

Off-policy TD learning cannot guarantee convergence and may even diverge in the classic Baird's counterexample (Baird et al., 1995). To ensure convergence, Sutton et al. (2009) proposed minimizing the mean squared projected Bellman error (MSPBE) as the optimization objective,

$$\text{MSPBE}(\boldsymbol{\theta}) = \|\Pi \mathcal{T}^\pi \mathbf{V}_\theta - \mathbf{V}_\theta\|_{\mathbf{D}}^2,$$

and derived the TD with gradient correction (TDC) algorithm. The update rule of TDC is

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha_t [\rho_t \delta_t \boldsymbol{\phi}_t - \gamma \boldsymbol{\phi}_{t+1} (\boldsymbol{\phi}_t^\top \mathbf{u}_t)], \quad (9)$$

where

$$\mathbf{u}_{t+1} = \mathbf{u}_t + \zeta_t [\rho_t \delta_t - \boldsymbol{\phi}_t^\top \mathbf{u}_t] \boldsymbol{\phi}_t. \quad (10)$$

2.2. Reward centering

Naik et al. (2024) proposed two reward centering algorithms, namely simple reward centering and value-based reward centering. The update rule of the simple reward centering is

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t \bar{\delta}_t, \quad (11)$$

where the new TD error $\bar{\delta}_t$ is

$$\bar{\delta}_t = \delta_t - \bar{r}_t = r_{t+1} - \bar{r}_t + \gamma V_t(s_{t+1}) - V_t(s_t), \quad (12)$$

and $\bar{r}_t$ is updated as:

$$\bar{r}_{t+1} = \bar{r}_t + \beta_t (r_{t+1} - \bar{r}_t), \quad (13)$$

where $\beta_t \in (0, 1)$ is a learning stepsize.

The update rule of the value-based reward centering is

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t \bar{\delta}_t, \quad (14)$$

where $\bar{r}_t$ is updated as:

$$\bar{r}_{t+1} = \bar{r}_t + \beta_t \bar{\delta}_t. \quad (15)$$

3. Bellman error centering

Given a vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ and its distribution $\mathbf{d} = (d_1, d_2, \dots, d_n)^\top$, where $\sum_{i=1}^n d_i = 1$, and $0 \leq d_i \leq 1$. Centering operator C is defined as follows:

$$C\mathbf{x} = \mathbf{x} - \mathbb{E}[\mathbf{x}] \mathbf{1} = \mathbf{x} - \sum_{i=1}^n x_i \mathbf{1} = \mathbf{x} - \mathbf{x}^\top \mathbf{d} \mathbf{1}, \quad (16)$$

where $\mathbf{1}$ is an all-ones vector. For each element $x_i \in \mathbf{x}$, $Cx_i = x_i - \mathbb{E}[\mathbf{x}] \mathbf{1}$. For any vector $\mathbf{x}$ and $\mathbf{y}$ with a same distribution $\mathbf{d}$, we have

$$\begin{aligned} C(\mathbf{x} + \mathbf{y}) &= (\mathbf{x} + \mathbf{y}) - (\mathbf{x} + \mathbf{y})^\top \mathbf{d} \mathbf{1} \\ &= \mathbf{x} - \mathbf{x}^\top \mathbf{d} \mathbf{1} + \mathbf{y} - \mathbf{y}^\top \mathbf{d} \mathbf{1} \\ &= C\mathbf{x} + C\mathbf{y}. \end{aligned} \quad (17)$$

3.1. Revisit reward centering

Let's revisit the two reward centering algorithms and analyze their updating rules using the idea of two-time scales. For simple reward centering, assume that recursion (11) is the slow-time scale and recursion (13) is the fast-time scale. Therefore, the recursion (13) will approach the average reward,

$$\bar{r}_t \approx \mathbb{E}[r_t]. \quad (18)$$

That is $r_t - \bar{r}_t \approx r_t - \mathbb{E}[r_t] = Cr_t$. Then, the simple reward centering can be rewritten as:

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t [Cr_{t+1} + \gamma V_t(s_{t+1}) - V_t(s_t)]. \quad (19)$$

Therefore, the simple reward centering is, in a strict sense, reward centering.

By definition of $\bar{\delta}_t = \delta_t - \bar{r}_t$, let's rewrite the update rule of the value-based reward centering as follows:

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t (\delta_t - \bar{r}_t), \quad (20)$$

where $\bar{r}_t$ is updated as:

$$\bar{r}_{t+1} = \bar{r}_t + \beta_t (\delta_t - \bar{r}_t). \quad (21)$$

Similarly, assume that recursion (20) is the slow-time scale and recursion (21) is the fast-time scale. Therefore, the recursion (21) will approach the expected Bellman error.

$$\bar{r}_t \approx \mathbb{E}[\delta_t]. \quad (22)$$

That is $\delta_t - \bar{r}_t \approx C\delta_t$. Then, the value-based reward centering can be rewritten as:

$$V_{t+1}(s_t) = V_t(s_t) + \alpha_t C\delta_t. \quad (23)$$

Therefore, the value-based reward centering is no more, in a strict sense, reward centering. It is, in a strict sense, **Bellman error centering**.

It is worth noting that this understanding is crucial, as designing new algorithms requires leveraging this concept.

3.2. On the fixed point solution

The update rule (23) is a stochastic approximation of the following update:

$$\begin{aligned} V_{t+1} &= V_t + \alpha_t [\mathcal{T}^\pi V - V - \mathbb{E}[\delta] \mathbf{1}] \\ &= V_t + \alpha_t [\mathcal{T}^\pi V - V - (\mathcal{T}^\pi V - V)^\top d_\pi \mathbf{1}] \\ &= V_t + \alpha_t [C(\mathcal{T}^\pi V - V)]. \end{aligned} \quad (24)$$

If update rule (24) converges, it is expected that $C(\mathcal{T}^\pi V - V) = \mathbf{0}$. That is

$$\begin{aligned} CV &= C\mathcal{T}^\pi V \\ &= C(R^\pi + \gamma \mathbb{P}^\pi V) \\ &= CR^\pi + \gamma C\mathbb{P}^\pi V \\ &\quad (\text{The distributions of } R^\pi \text{ and } \mathbb{P}^\pi V \text{ are the same.}) \\ &= CR^\pi + \gamma(\mathbb{P}^\pi V - (\mathbb{P}^\pi V)^\top d_\pi \mathbf{1}) \\ &= CR^\pi + \gamma(\mathbb{P}^\pi V - V^\top (\mathbb{P}^\pi)^\top d_\pi \mathbf{1}) \\ &\quad ((\mathbb{P}^\pi)^\top d_\pi = d_\pi.) \\ &= CR^\pi + \gamma(\mathbb{P}^\pi V - V^\top d_\pi \mathbb{P}^\pi \mathbf{1}) \\ &\quad (\mathbf{1} = \mathbb{P}^\pi \mathbf{1}.) \\ &= CR^\pi + \gamma(\mathbb{P}^\pi V - \mathbb{P}^\pi V^\top d_\pi \mathbf{1}) \\ &\quad (V^\top d_\pi \text{ is a scalar.}) \\ &= CR^\pi + \gamma \mathbb{P}^\pi (V - V^\top d_\pi \mathbf{1}) \\ &= CR^\pi + \gamma \mathbb{P}^\pi CV \\ &\doteq \mathcal{T}_c^\pi CV \end{aligned} \quad (25)$$

where we defined $\mathcal{T}_c^\pi$ as a centered Bellman operator. We call Eq. (25) centered Bellman equation. And it is **centered fixed point**.

For linear value function approximation, let's define

$$\begin{aligned} \Pi \mathcal{T}_c^\pi CV_\theta &= \Pi CV_\theta \\ &= \Pi(V_\theta - V_\theta^\top d_\pi \mathbf{1}) \\ &= \Pi V_\theta - V_\theta^\top d_\pi \Pi \mathbf{1} \\ &\quad (V_\theta^\top d_\pi \text{ is a scalar.}) \\ &= V_\theta - V_\theta^\top d_\pi \mathbf{1} \\ &\quad (\text{Both } V_\theta \text{ and } \mathbf{1} \text{ are in the feature space.}) \\ &\doteq CV_\theta \end{aligned} \quad (26)$$

We call Eq. (26) **centered TD fixed point**.

3.3. Centered TD algorithms

Given the above centered TD fixed point, mean squared centered Bellman error (MSCBE), is proposed as follows:

$$\begin{aligned} \arg \min_{\theta} \text{MSCBE}(\theta) &= \arg \min_{\theta} \|\mathcal{T}_c^\pi CV_\theta - CV_\theta\|_D^2 \\ &= \arg \min_{\theta} \|\mathcal{T}^\pi V_\theta - V_\theta - (\mathcal{T}^\pi V_\theta - V_\theta)^\top d \mathbf{1}\|_D^2 \\ &= \arg \min_{\theta, \omega} \|\mathcal{T}^\pi V_\theta - V_\theta - \omega \mathbf{1}\|_D^2. \end{aligned}$$

where ω is used to estimate the expected value of the Bellman error.

First, the parameter ω is derived directly based on stochastic gradient descent:

$$\omega_{t+1} = \omega_t + \beta_t (\delta_t - \omega_t). \quad (27)$$

Then, to obtain the centered TD fixed point, we optimize θ using the semi-gradient method.,

$$\theta_{t+1} = \theta_t + \alpha_t (\delta_t - \omega_t) \phi_t. \quad (28)$$

We call (27) and (28) on-policy centered TD (CTD) algorithm. The convergence analysis of on-policy CTD will be given in the following section.

Update rules of off-policy CTD are then,

$$\omega_{t+1} = \omega_t + \beta_t (\rho_t \delta_t - \omega_t), \quad (29)$$

$$\theta_{t+1} = \theta_t + \alpha_t (\rho_t \delta_t - \omega_t) \phi_t. \quad (30)$$

3.4. Off-policy centered TDC algorithm

The convergence of the off-policy centered TD algorithm may not be guaranteed.

To deal with this problem, we propose another new objective function, called mean squared projected centered Bellman error (MSPCBE), and derive Centered TDC algorithm (CTDC).

The specific expression of the objective function MSPCBE is as follows:

$$\begin{aligned} \arg \min_{\theta} \text{MSPCBE}(\theta) &= \arg \min_{\theta} \|\Pi \mathcal{T}_c^\pi CV_\theta - CV_\theta\|_D^2 \\ &= \arg \min_{\theta} \|\Pi(\mathcal{T}_c^\pi CV_\theta - CV_\theta)\|_D^2 \\ &= \arg \min_{\theta, \omega} \|\Pi(\mathcal{T}^\pi V_\theta - V_\theta - \omega \mathbf{1})\|_D^2. \end{aligned}$$

The derivation process of CTDC is the same as for the TDC algorithm (Sutton et al., 2009), the only difference is that the original δ is replaced by $\delta - \omega$. The updated rules of CTDC algorithm are as follows:

$$\theta_{t+1} = \theta_t + \alpha_t [(\delta_t - \omega_t) \phi_t - \gamma \phi_{t+1} (\phi_t^\top u_t)], \quad (31)$$

$$u_{t+1} = u_t + \zeta_t [\delta_t - \omega_t - \phi_t^\top u_t] \phi_t, \quad (32)$$

and

$$\omega_{t+1} = \omega_t + \beta_t (\delta_t - \omega_t). \quad (33)$$

where u is an updated parameter. This algorithm is derived to work with a given set of sub-samples—in the form of triples (S_t, R_t, S'_t) that match transitions from both the behavior and target policies.

4. Convergence analysis

The purpose of this section is to establish the convergence guarantees of CTD and CTDC under standard assumptions, as well as the invariance of the optimal policy.

4.1. Convergence proof of on-policy CTD

Theorem 1. (Convergence of on-policy CTD). In the case of on-policy learning, consider the iterations (27) and (28). Let the step-size sequences α_k and β_k , $k \geq 0$ where k is the iteration step satisfy in this case $\alpha_k, \beta_k > 0$, for all k , $\sum_{k=0}^{\infty} \alpha_k = \sum_{k=0}^{\infty} \beta_k = \infty$, $\sum_{k=0}^{\infty} \alpha_k^2 < \infty$, $\sum_{k=0}^{\infty} \beta_k^2 < \infty$, and $\alpha_k = o(\beta_k)$. Assume that (ϕ_k, r_k, ϕ'_k) is an i.i.d. sequence with uniformly bounded second moments, where ϕ_k and ϕ'_k are sampled from the same Markov chain. Let $\mathbf{A} = \text{Cov}(\phi, \phi - \gamma\phi')$, $\mathbf{b} = \text{Cov}(r, \phi)$. Assume that matrix $\mathbf{A}$ is non-singular. Then the parameter vector θ_k converges with probability one to the centered TD fixed point $\mathbf{A}^{-1}\mathbf{b}$.

Proof. The proof is based on Borkar's Theorem for general stochastic approximation recursions with two time scales (Borkar, 1997).

The centered one-step linear TD solution is defined as:

$$0 = \mathbb{E}[(\delta - \mathbb{E}[\delta])\phi] = -\mathbf{A}\theta + \mathbf{b}.$$

Thus, the on-policy CTD algorithm's solution is $\theta_{\text{CTD}} = \mathbf{A}^{-1}\mathbf{b}$. This solution is equivalent to the fixed point (26).

First, note that recursion (28) can be rewritten as

$$\theta_{k+1} \leftarrow \theta_k + \beta_k \xi(k),$$

where

$$\xi(k) = \frac{\alpha_k}{\beta_k} (\delta_k - \omega_k) \phi_k.$$

Due to the settings of step-size schedule $\alpha_k = o(\beta_k)$, $\xi(k) \rightarrow 0$ almost surely as $k \rightarrow \infty$. That is the increments in iteration (27) are uniformly larger than those in (28), thus (27) is the faster recursion. Along the faster timescale, iterations of (27) and (28) are associated with the ODEs system as follows:

$$\dot{\theta}(t) = 0, \quad (34)$$

$$\dot{\omega}(t) = \mathbb{E}[\delta_t | \theta(t)] - \omega(t). \quad (35)$$

Based on the ODE (34), $\theta(t) \equiv \theta$ when viewed from the faster timescale. By Hirsch lemma (Hirsch, 1989), it follows that $\|\theta_k - \theta\| \rightarrow 0$ a.s. as $k \rightarrow \infty$ for some θ that depends on the initial condition θ_0 of recursion (28). Thus, the ODE pair (34)-(35) can be written as

$$\dot{\omega}(t) = \mathbb{E}[\delta_t | \theta] - \omega(t). \quad (36)$$

Consider the function $h(\omega) = \mathbb{E}[\delta | \theta] - \omega$, i.e., the driving vector field of the ODE (36). It is easy to find that the function h is Lipschitz with coefficient -1 . Let $h_{\infty}(\cdot)$ be the function defined by $h_{\infty}(\omega) = \lim_{x \rightarrow \infty} \frac{h(x\omega)}{x}$. Then $h_{\infty}(\omega) = -\omega$, is well-defined. For (36), $\omega^* = \mathbb{E}[\delta | \theta]$ is the unique globally asymptotically stable equilibrium. For the ODE

$$\dot{\omega}(t) = h_{\infty}(\omega(t)) = -\omega(t), \quad (37)$$

apply $\bar{V}(\omega) = (-\omega)^T(-\omega)/2$ as its associated strict Lyapunov function. Then, the origin of (37) is a globally asymptotically stable equilibrium.

Consider now the recursion (27). Let $M_{k+1} = (\delta_k - \omega_k) - \mathbb{E}[(\delta_k - \omega_k) | \mathcal{F}(k)]$, where M is the martingale, $\mathcal{F}(k) = \sigma(\omega_l, \theta_l, l \leq k; \phi_s, \phi'_s, r_s, s < k)$, $k \geq 1$ are the sigma fields generated by $\omega_0, \theta_0, \omega_{l+1}, \theta_{l+1}, \phi_l, \phi'_l$, $0 \leq l < k$. It is easy to verify that $M_{k+1}, k \geq 0$ are integrable random variables that satisfy $\mathbb{E}[M_{k+1} | \mathcal{F}(k)] = 0$, $\forall k \geq 0$. Because ϕ_k, r_k , and ϕ'_k have uniformly bounded second moments, it can be seen that for some constant $c_1 > 0$, $\forall k \geq 0$,

$$\mathbb{E}[\|M_{k+1}\|^2 | \mathcal{F}(k)] \leq c_1(1 + \|\omega_k\|^2 + \|\theta_k\|^2).$$

Now Assumptions (A1) and (A2) of Borkar and Meyn (2000) are verified. Furthermore, Assumptions (TS) of Borkar and Meyn (2000) are satisfied by our conditions on the step-size sequences α_k, β_k . Thus, by

Theorem 2.2 of Borkar and Meyn (2000) we obtain that $\|\omega_k - \omega^*\| \rightarrow 0$ almost surely as $k \rightarrow \infty$.

Consider now the slower timescale recursion (28). Based on the above analysis, (28) can be rewritten as

$$\theta_{k+1} \leftarrow \theta_k + \alpha_k (\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k.$$

Let $\mathcal{G}(k) = \sigma(\theta_l, l \leq k; \phi_s, \phi'_s, r_s, s < k)$, $k \geq 1$ be the sigma fields generated by $\theta_0, \theta_{l+1}, \phi_l, \phi'_l$, $0 \leq l < k$. Let $Z_{k+1} = Y_k - \mathbb{E}[Y_k | \mathcal{G}(k)]$, where

$$Y_k = (\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k.$$

Consequently,

$$\begin{aligned} \mathbb{E}[Y_k | \mathcal{G}(k)] &= \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k | \mathcal{G}(k)] \\ &= \mathbb{E}[\delta_k \phi_k | \theta_k] - \mathbb{E}[\mathbb{E}[\delta_k | \theta_k] \phi_k] \\ &= \mathbb{E}[\delta_k \phi_k | \theta_k] - \mathbb{E}[\delta_k | \theta_k] \mathbb{E}[\phi_k] \\ &= \text{Cov}(\delta_k | \theta_k, \phi_k), \end{aligned}$$

where $\text{Cov}(\cdot, \cdot)$ is a covariance operator.

Thus,

$$Z_{k+1} = (\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k - \text{Cov}(\delta_k | \theta_k, \phi_k).$$

It is easy to verify that $Z_{k+1}, k \geq 0$ are integrable random variables that satisfy $\mathbb{E}[Z_{k+1} | \mathcal{G}(k)] = 0$, $\forall k \geq 0$. Also, because ϕ_k, r_k , and ϕ'_k have uniformly bounded second moments, it can be seen that for some constant $c_2 > 0$, $\forall k \geq 0$,

$$\mathbb{E}[\|Z_{k+1}\|^2 | \mathcal{G}(k)] \leq c_2(1 + \|\theta_k\|^2).$$

Consider now the following ODE associated with (28):

$$\begin{aligned} \dot{\theta}(t) &= \text{Cov}(\delta | \theta(t), \phi) \\ &= \text{Cov}(r + (\gamma\phi' - \phi)^\top \theta(t), \phi) \\ &= \text{Cov}(r, \phi) - \text{Cov}(\theta(t)^\top (\phi - \gamma\phi'), \phi) \\ &= \text{Cov}(r, \phi) - \theta(t)^\top \text{Cov}(\phi - \gamma\phi', \phi) \\ &= \text{Cov}(r, \phi) - \text{Cov}(\phi - \gamma\phi', \phi)^\top \theta(t) \\ &= \text{Cov}(r, \phi) - \text{Cov}(\phi, \phi - \gamma\phi') \theta(t) \\ &= -\mathbf{A}\theta(t) + \mathbf{b}. \end{aligned} \quad (38)$$

Let $h(\theta(t))$ be the driving vector field of the ODE (38).

$$h(\theta(t)) = -\mathbf{A}\theta(t) + \mathbf{b}.$$

Consider the cross-covariance matrix,

$$\begin{aligned} \mathbf{A} &= \text{Cov}(\phi, \phi - \gamma\phi') \\ &= \frac{\text{Cov}(\phi, \phi) + \text{Cov}(\phi - \gamma\phi', \phi - \gamma\phi') - \text{Cov}(\gamma\phi', \gamma\phi')}{2} \\ &= \frac{\text{Cov}(\phi, \phi) + \text{Cov}(\phi - \gamma\phi', \phi - \gamma\phi') - \gamma^2 \text{Cov}(\phi', \phi')}{2} \\ &= \frac{(1 - \gamma^2) \text{Cov}(\phi, \phi) + \text{Cov}(\phi - \gamma\phi', \phi - \gamma\phi')}{2}, \end{aligned} \quad (39)$$

where we eventually used $\text{Cov}(\phi', \phi') = \text{Cov}(\phi, \phi)$ ¹. Note that the covariance matrix $\text{Cov}(\phi, \phi)$ and $\text{Cov}(\phi - \gamma\phi', \phi - \gamma\phi')$ are semi-positive definite. Then, the matrix $\mathbf{A}$ is semi-positive definite because $\mathbf{A}$ is linearly combined by two positive-weighted semi-positive definite matrices. Furthermore, $\mathbf{A}$ is nonsingular due to the assumption. Hence, the cross-covariance matrix $\mathbf{A}$ is positive definite.

Therefore, $\theta^* = \mathbf{A}^{-1}\mathbf{b}$ can be seen to be the unique globally asymptotically stable equilibrium for ODE (38). Let $h_{\infty}(\theta) = \lim_{r \rightarrow \infty} \frac{h(r\theta)}{r}$. Then $h_{\infty}(\theta) = -\mathbf{A}\theta$ is well-defined. Consider now the ODE

$$\dot{\theta}(t) = -\mathbf{A}\theta(t). \quad (40)$$

The ODE (40) has the origin as its unique globally asymptotically stable equilibrium. Thus, the assumption (A1) and (A2) are verified. $\square$

¹ The covariance matrix $\text{Cov}(\phi', \phi')$ is equal to the covariance matrix $\text{Cov}(\phi, \phi)$ if the initial state is re-reachable or initialized randomly in a Markov chain for on-policy update.

4.2. Convergence proof of off-policy CTDC

Theorem 2. (Convergence of off-policy CTDC). In the case of off-policy learning, consider the iterations (33), (32) and (31). Let the step-size sequences α_k , ζ_k and β_k , $k \geq 0$ satisfy in this case $\alpha_k, \zeta_k, \beta_k > 0$, for all k , $\sum_{k=0}^{\infty} \alpha_k = \sum_{k=0}^{\infty} \beta_k = \sum_{k=0}^{\infty} \zeta_k = \infty$, $\sum_{k=0}^{\infty} \alpha_k^2 < \infty$, $\sum_{k=0}^{\infty} \zeta_k^2 < \infty$, $\sum_{k=0}^{\infty} \beta_k^2 < \infty$, and $\alpha_k = o(\zeta_k)$, $\zeta_k = o(\beta_k)$. Assume that (ϕ_k, r_k, ϕ'_k) is an i.i.d. sequence with uniformly bounded second moments. Let $\mathbf{A} = \text{Cov}(\phi, \phi - \gamma\phi')$, $\mathbf{b} = \text{Cov}(r, \phi)$, and $\mathbf{C} = \mathbb{E}[\phi\phi^\top]$. Assume that $\mathbf{A}$ and $\mathbf{C}$ are non-singular matrices. Then the parameter vector θ_k converges with probability one to the centered TD fixed point $\mathbf{A}^{-1}\mathbf{b}$ (26).

Proof. The proof is similar to that given by Sutton et al. (2009) for TDC, but it is based on multi-time-scale stochastic approximation.

A sketch proof is given as follows. In the fastest timescale, the parameter w converges to $\mathbb{E}[\delta|u_k, \theta_k]$. In the second fast timescale, the parameter u converges to $\mathbf{C}^{-1}\mathbb{E}[(\delta - \mathbb{E}[\delta|\theta_k])\phi|\theta_k]$. In the slower timescale, the associated ODE is

$$\dot{h}(\theta(t)) = \mathbf{A}^\top \mathbf{C}^{-1}(-\mathbf{A}\theta(t) + \mathbf{b}).$$

The matrix $\mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A}$ is positive definite. Thus, the parameter θ converges to $\mathbf{A}^{-1}\mathbf{b}$. $\square$

Please refer to Appendix A for CTDC's detailed proof.

4.3. On the quality of the solution

This section will discuss the optimal policy invariance of the CTD and CTDC algorithms.

Reward shaping can significantly speed up learning by adding a shaping reward $F(s, s')$ to the original reward r , where $F(s, s')$ is the general form of any state-based shaping reward. Static potential-based reward shaping (Static PBRS) maintains the policy invariance if the shaping reward follows from $F(s, s') = \gamma f(s') - f(s)$ (Ng et al., 1999).

This means that we can make changes to the TD error $\delta = r + \gamma\theta^\top \phi' - \theta^\top \phi$ while still ensuring the invariance of the optimal policy,

$$\delta - \omega = r + \gamma\theta^\top \phi' - \theta^\top \phi - \omega,$$

where ω is a constant, acting as a static PBRS. However, selecting an appropriate ω requires expert knowledge. This forces us to learn ω dynamically, i.e., $\omega = \omega_t$ and dynamic PBRS can maintain the policy invariance if the shaping reward is $F(s, t, s', t') = \gamma f(s', t') - f(s, t)$, where t is the time-step the agent reaches in state s (Devlin & Kudenko, 2012). However, this result requires the convergence guarantee of the dynamic potential function $f(s, t)$. If $f(s, t)$ does not converge as the time-step $t \rightarrow \infty$, the Q-values of dynamic PBRS are not guaranteed to converge.

Theorem 3. (Optimal policy invariance of CTD). Consider the iterations (27) and (28) of CTD. The CTD algorithm maintains the optimal policy invariance.

Proof. Consider the parameter variable

$$\omega_t = \gamma \left(\frac{\omega_t}{\gamma - 1} \right) - \left(\frac{\omega_t}{\gamma - 1} \right).$$

Let $\eta(s) \equiv \frac{1}{\gamma - 1}$, consequently,

$$f_{\omega_t}(s, t) = \omega_t^\top \zeta(s) = \frac{\omega_t}{\gamma - 1}$$

Thus, $F_{\omega_t}(s, s') = \gamma f_{\omega_t}(s') - f_{\omega_t}(s) = \omega_t$ is a dynamic potential-based function. $\square$

Similarly, the CTDC algorithm ensures optimal policy invariance.

5. Experimental studies

To evaluate the proposed CTD and CTDC algorithms, which are primarily designed for policy evaluation, we conduct comprehensive experiments comparing them against their baseline counterparts: TD and TDC.

To validate the convergence guarantees established in Section 4, we prioritize testing on classic counterexamples where traditional methods diverge, i.e., 2-state counterexample and Baird's counterexample, to observe empirical convergence behavior.

Furthermore, our work introduces novel centered fixed points (centered Bellman fixed point and centered TD fixed point), which differ from conventional fixed points, while preserving optimal policy invariance. Ultimately, policy evaluation serves policy improvement in control tasks; thus, we extend TD, TDC, CTD, and CTDC to learning control settings. For intuitive comparison, we first eliminate function approximation (a component of the deadly triad) by using tabular value functions, ensuring convergence across all methods, and test on Maze and Cliff Walking environments. Subsequently, we incorporate linear function approximation and evaluate on Mountain Car and Cart Pole tasks (Chen et al., 2023a). Finally, to assess algorithmic robustness, we perform parameter sensitivity analyses across varying hyperparameters.

5.1. Experimental settings

2-state counterexample (Sutton et al., 2016): Each state in 2-state counterexample, see Fig. 1(a), has two actions, *left* and *right*, which take the agent to the left or the right state. All rewards are zero. The feature $\Phi = (1, 2)^\top$ are assigned to the left and the right state. The behavior policy takes equal probability to the left or the right in both states, i.e., $\mathbf{P}_\mu = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$. The target policy only selects action *right* in both states, i.e., $\mathbf{P}_\pi = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$. The state distribution of the behavior policy is $\mathbf{d}_\mu = (0.5, 0.5)^\top$. The discount factor is $\gamma = 0.9$.

Baird's counterexample (Baird et al., 1995; Sutton et al., 2009): There are 7 states in Baird's counterexample, see Fig. 1(b). The behavior policy chooses actions represented by solid lines with a probability of $\frac{1}{7}$ and actions represented by dotted lines with a probability of $\frac{6}{7}$. The target policy chooses the solid line with the probability of 1. Rewards for each transition are zero. Thus, the true values are zeros for all states and for any given policy. The discount factor $\gamma = 0.99$. The feature matrix is:

$$\Phi = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Maze: In maze, see Fig. 1(c), the objective is to find the shortest path from the upper-left corner to the lower-right corner. In each state, there are four alternative actions: *up*, *down*, *left*, and *right*, which takes the agent deterministically to the corresponding neighbor state, except when a movement is blocked by an obstacle or the edge of the maze. Rewards are -1 in all transitions until the agent reaches the goal state. The discount factor $\gamma = 0.99$. The maximum number of steps in the episode is set to 1000.

Cliff Walking, Mountain Car, and Acrobot: They are selected from the gym's official website and correspond to the following versions: "CliffWalking-v0", "MountainCar-v0" and "Acrobot-v1". For specific details, please refer to the gym's official website. The maximum number of steps for the Mountain Car environment is set to 1000, while the default settings are used for the other two environments. In the Mountain Car and Acrobot tasks, tile coding is adopted for feature extraction in linear value function approximation, with 5 tilings configured for each task. Specifically, each tiling corresponds to a 20×20 feature grid in Mountain Car, while each tiling corresponds to a $5 \times 5 \times 5 \times 5 \times 5$ high-dimensional feature grid in Acrobot.

In control tasks, the behavior policy of each algorithm adopts the ϵ -greedy policy, while the target policy is the greedy policy; the initial

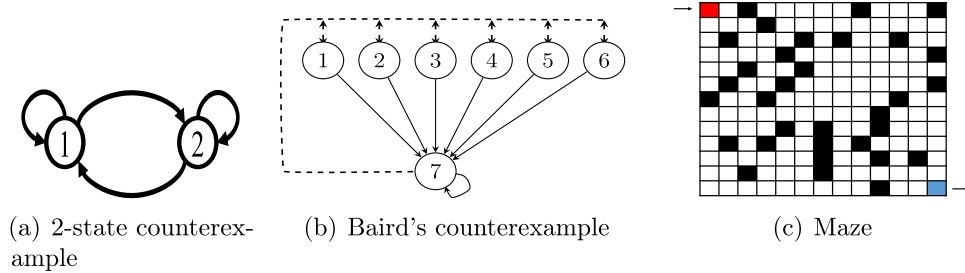


Fig. 1. Counterexamples and Maze.

Table 1

Parameter settings of various algorithms in different environments (TD: α ; CTD: α & β ; TDC: α & ζ ; CTDC: α , β & ζ).

Environment	α	β	ζ
2-state	0.1	0.0001	0.001
Baird's	0.01	0.01	0.05
Maze	0.1	0.001	0.001
Cliff Walking	0.1	0.0001	0.005
Mountain Car	0.1	0.0001	0.001
Acrobot	0.1	0.0001	0.1

Table 2

Parameters for policy evaluation.

Counterexample	Parameters	Values
2-State	α	{0.0001, 0.0005, 0.001, 0.005, 0.01}
	ζ	{0.0005, 0.001, 0.005, 0.01, 0.05}
	β	{0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.2}
Baird's	α	{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3}
	ζ	{0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.2}
	β	{0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.2}

value of ϵ is set to 0.01, which decays by a factor of 0.9992 per episode and is constrained to be no less than 0.0001.

To ensure the fairness of experimental comparisons, we uniformly set the parameters of the corresponding types for each algorithm to the same values; detailed settings of the learning rate parameters for each algorithm across different environments are presented in Table 1. In the CTD and CTDC algorithms, the value of the learning rate parameter β should not be set too small; otherwise, the error term cannot be effectively learned, which will in turn cause the performance of the two algorithms to approach that of the TD algorithm and TDC algorithm respectively. Meanwhile, its value should not be too large either, because the parameter ω is a scalar rather than a vector and needs to be updated at each step.

5.2. Experimental results analysis

All experiments are independently run 50 times to ensure statistical reliability, with results reported as means and standard deviations; shaded regions in figures represent variance for visual clarity.

Fig. 2 presents the learning curves of the four algorithms in prediction and control tasks, and their performance can be analyzed separately from the perspectives of prediction and control tasks.

In prediction tasks, first, from the results of the 2-state counterexample and Baird's counterexample in Fig. 2(a) and (b), the TD algorithm exhibits a divergent trend, while the TDC algorithm achieves stable convergence, which is fully consistent with the conclusions of existing studies (Sutton et al., 2009). Second, both the CTD and CTDC algorithms achieve convergence: the convergence of CTDC is as expected and consistent with the theoretical derivation of Theorem 2; while the convergence of CTD is rather unexpected, since its convergence can only be theoretically guaranteed in on-policy learning scenarios, and there is no clear theoretical support for its convergence in off-policy scenarios. Further interpretation and analysis of the convergence of CTD in the above two experiments will be provided in the next section. Finally, in the 2-state counterexample, the convergence speed of CTD and CTDC is significantly faster than that of TDC; however, in Baird's counterexample, their convergence speed is slightly slower than that of TDC, indicating that the three improved algorithms have their own advantages and disadvantages in different scenarios.

In control tasks, in the maze and cliff walking tasks using tabular value functions (Fig. 2(c), (d)), all algorithms eventually converge to the

same optimal solution, which verifies the effectiveness of the proposed new fixed-point theory and invariance of the optimal policy. Meanwhile, the convergence speed of CTD and CTDC algorithms is significantly faster than that of TD and TDC algorithms. In the mountain car and acrobot tasks using tile coding value function approximation (Fig. 2(e), (f)), CTD and CTDC algorithms not only converge faster but also achieve significantly better final convergence accuracy than TD and TDC algorithms, which fully demonstrates that the error centralization idea has more prominent advantages in algorithm performance.

5.3. Parameter sensitivity experiments

To verify the hyperparameter robustness of the proposed CTD and CTDC algorithms and quantify the impact of core parameters on traditional baselines (TD and TDC), we conduct multidimensional sensitivity tests on two classic non-convergent scenarios (2-state counterexample and Baird's counterexample). The hyperparameter candidates involved in the experiments are summarized in Table 2, and we adopt the normalized mean squared centered Bellman error (RMSCBE) as the core evaluation metric to measure the convergence, stability, and error magnitude of each algorithm under different parameter configurations.

In the 2-state counterexample, the baseline TD algorithm shows extreme sensitivity to the base learning rate α . As shown in Fig. 3, TD diverges rapidly when $\alpha \geq 0.01$, and can only maintain low RMSCBE within an extremely narrow range of $\alpha \leq 0.001$, which is consistent with the theoretical conclusion that the TD matrix is negative definite in this scenario (Section 6.1). For TDC, the introduction of gradient correction parameter ζ improves stability to a certain extent, but excessive ζ (greater than 0.05) will cause the correction term to over-perturb the parameters, leading to a rebound in RMSCBE, reflecting the strict constraints on ζ for TDC. In contrast, the proposed CTD algorithm maintains stable RMSCBE (below 0.2) within a wide range of β values (0.0005 to 0.1) without divergence; only when $\beta > 0.2$ will the scalar parameter ω update too fast and trigger fluctuations, which verifies the parameter constraint hypothesis in Section 5.1. The CTDC algorithm achieves stable convergence (RMSCBE < 0.51) under different baseline α values by adjusting the combination of β and ζ , and there is no divergence risk even when α increases to 0.1, proving the strong adaptability of its multi-parameter collaborative mechanism (Fig. 4).

In Baird's counterexample, the baseline TD algorithm shows high RMSCBE (greater than 3.0) across all α ranges, and the error exceeds 4.5 when $\alpha > 0.01$, reconfirming the theoretical defect that the TD ma-

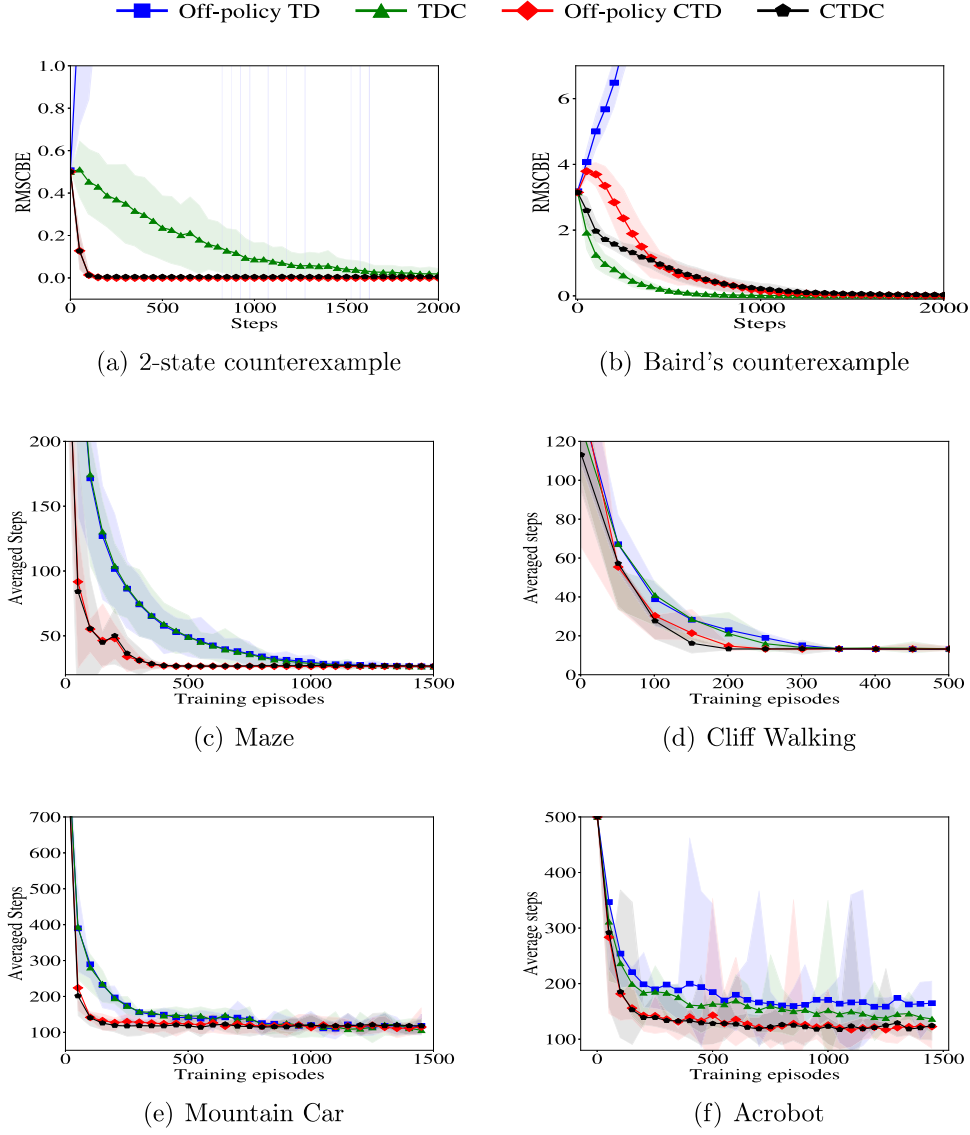


Fig. 2. Learning curves in policy evaluation and control tasks.

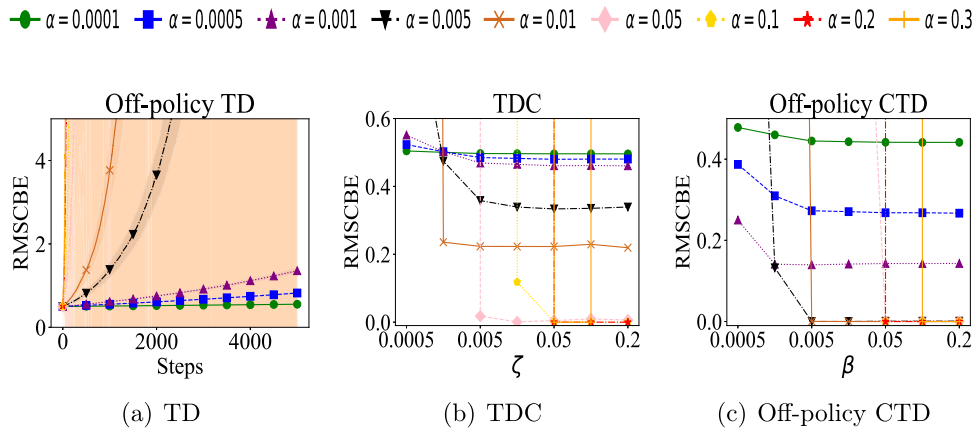


Fig. 3. Sensitivity of TD, TDC and off-policy CTD to learning rates in 2-state counterexample.

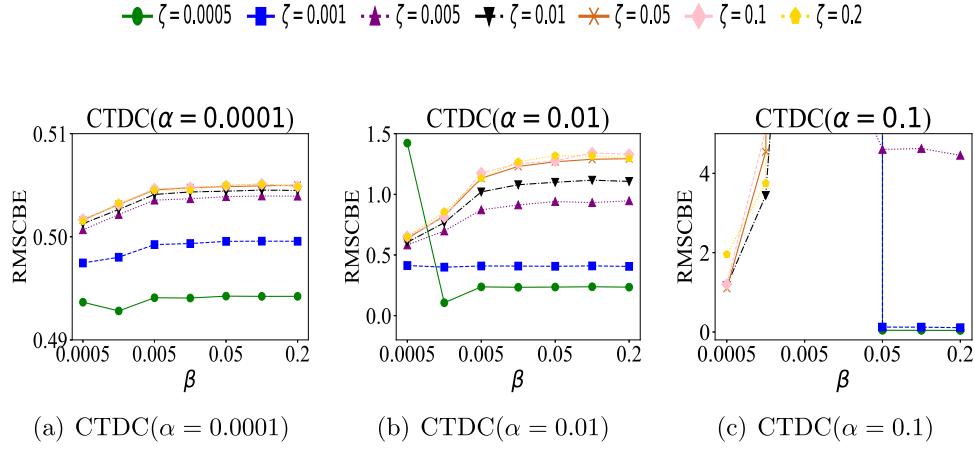
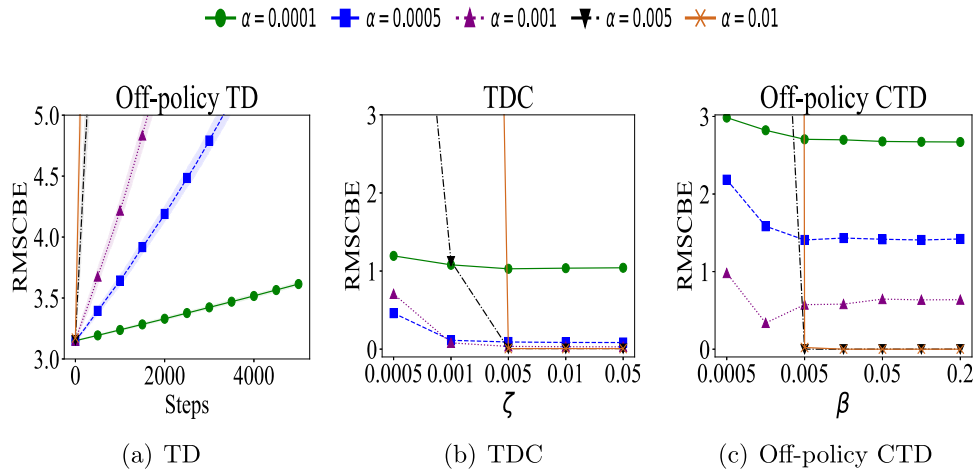
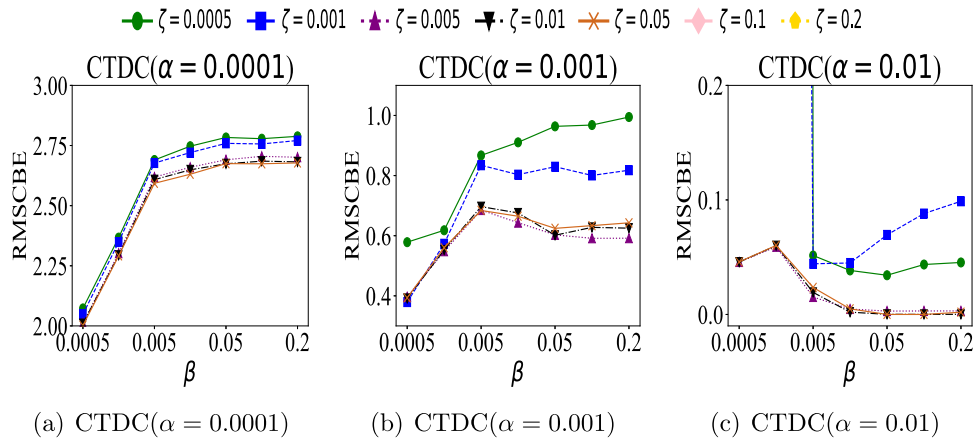
Fig. 4. Sensitivity of CTDC with different α to learning rates in 2-state counterexample.

Fig. 5. Sensitivity of off-policy TD, TDC and off-policy CTD to learning rates in Baird's counterexample.

Fig. 6. Sensitivity of CTDC with different α to learning rates in Baird's counterexample.

trix is non-positive definite in this scenario (Fig. 5). The TDC algorithm can only control RMSCE below 3.0 within an extremely narrow parameter interval, showing poor adaptability to high-dimensional feature scenarios. For the proposed algorithms, CTD maintains RMSCE below 2.0 when β is in the range of 0.0005 to 0.01, and the effective interval of α is expanded by 10 times compared with TDC; even when $\beta = 0.1$, there is no obvious divergence, which is consistent with the conclusion that the CTD matrix is positive semi-definite in Section 6.2. The CTDC

algorithm can reduce RMSCE to below 0.2 under the optimal parameter combination, and still maintains a convergent state with error < 1.0 when $\alpha = 0.01$, which fully demonstrates that the collaborative mechanism of error centering and gradient correction can effectively reduce parameter sensitivity in complex scenarios (Fig. 6).

Three key findings can be drawn from the sensitivity experiments. First, the parameter robustness follows the hierarchy of CTDC > CTD > TDC > TD, and the error centering mechanism of the proposed al-

gorithms greatly broadens the effective parameter interval, solving the parameter sensitivity problem of traditional off-policy algorithms. Second, the optimal interval of β for CTD and CTDC is $[0.0005, 0.01]$; when $\beta < 0.0005$, the error term cannot be learned effectively and the performance approaches that of TD/TDC, while when $\beta > 0.1$, excessive updates of scalar ω will cause fluctuations, which is consistent with the parameter constraints in Section 5.1. Third, the three parameters of CTDC (base step size α , error centering β , gradient correction ζ) have complementary effects, and stable convergence can be achieved in complex scenarios through reasonable combination, while such synergy is not observed between α and ζ in TDC.

6. Numerical studies on off-policy TD and off-policy CTD

In this work, we have re-examined reward centering methodologies and established the Bellman Error Centering (BEC) framework as the mathematical foundation underlying Value-based Reward Centering (VRC). This perspective enabled us to derive novel theoretical results, including centered fixed-point solutions for both tabular value functions and linear function approximation. To operationalize these insights, we introduced two algorithms: the on-policy Centered Temporal Difference (CTD) algorithm and the off-policy Centered Temporal Difference with Correction (CTDC) algorithm. Formal convergence proofs were provided for both under standard assumptions, demonstrating their reliability in respective settings. Experimental evaluations across diverse benchmark environments highlighted the superior stability, faster convergence, and enhanced performance of our centered approaches compared to traditional TD and TDC baselines.

A key motivation in our algorithm design was addressing the limitations of CTD in off-policy learning. While CTD is theoretically guaranteed to converge in on-policy scenarios, we lack a formal proof for its convergence in off-policy contexts due to potential issues with variance amplification and unbounded errors arising from importance sampling ratios. To mitigate this, we developed CTDC, which incorporates a gradient correction term to ensure stability and convergence in off-policy settings, analogous to the role of the correction in TDC.

Despite the absence of theoretical guarantees for off-policy CTD, our empirical results across policy evaluation and control tasks consistently showed stable convergence behavior. To better understand this discrepancy, we conducted additional numerical analyses on known off-policy counterexamples where off-policy TD diverges (Chen et al., 2024).

The update rule of off-policy TD is

$$\begin{aligned}\theta_{t+1} &\doteq \theta_t + \rho_t \alpha_t (R_{t+1} + \gamma \theta_t^\top \phi_{t+1} - \theta_t^\top \phi_t) \phi_t \\ &= \theta_t + \alpha_t \left(\underbrace{\rho_t R_{t+1} \phi_t}_{\mathbf{b}_t} - \underbrace{\rho_t \phi_t (\phi_t - \gamma \phi_{t+1})^\top \theta_t}_{\mathbf{A}_t} \right).\end{aligned}\quad (41)$$

The expected update of the off-policy TD is

$$\theta_{t+1} = \theta_t + \alpha_t (\mathbf{b} - \mathbf{A} \theta_t), \quad (42)$$

where

$$\begin{aligned}\mathbf{A} &= \lim_{t \rightarrow \infty} \mathbb{E}[\mathbf{A}_t] \\ &= \lim_{t \rightarrow \infty} \mathbb{E}_\mu [\rho_t \phi_t (\phi_t - \gamma \phi_{t+1})^\top] \\ &= \sum_s d_\mu(s) \mathbb{E}_\mu [\rho_k \phi_k (\phi_k - \gamma \phi_{k+1})^\top | S_k = s] \\ &= \sum_s d_\mu(s) \mathbb{E}_\pi [\phi_k (\phi_k - \gamma \phi_{k+1})^\top | S_k = s] \\ &= \sum_s d_\mu(s) \phi(s) \left(\phi(s) - \gamma \sum_{s'} [\mathbf{P}_\pi]_{ss'} \phi(s') \right)^\top \\ &= \Phi^\top \mathbf{D}_\mu (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi,\end{aligned}\quad (43)$$

$$\text{and } \mathbf{b} = \lim_{t \rightarrow \infty} \mathbb{E}[\mathbf{b}_t] = \Phi^\top \mathbf{D}_\mu \mathbf{r}_\pi.$$

Consider the expected behavior of the off-policy CTD algorithm of (30) and (29), we assume that ω_t has converges to $\mathbb{E}_\mu[\rho_t \delta_t | \theta_t]$ in the faster timescale.

$$\begin{aligned}\theta_{t+1} &= \theta_t + \alpha_t \rho_t (R_{t+1} + \gamma \theta_t^\top \phi_{t+1} - \theta_t^\top \phi_t) \phi_t - \alpha_t \mathbb{E}_\mu[\rho_t \delta_t | \theta_t] \phi_t \\ &= \theta_t + \alpha_t \left\{ \underbrace{(\rho_t R_{t+1} - \mathbb{E}_\mu[\rho_t R_{t+1}]) \phi_t}_{\mathbf{b}_t} \right. \\ &\quad \left. - \underbrace{(\rho_t \phi_t (\phi_t - \gamma \phi_{t+1})^\top - \mathbb{E}_\mu[\rho_t (\phi_t - \gamma \phi_{t+1})^\top] \theta_t)_{\mathbf{A}_t}} \right\}\end{aligned}\quad (44)$$

Then, the expected update of the off-policy CTD (Sutton et al., 2016) is

$$\theta_{t+1} = \theta_t + \alpha_t (\mathbf{b} - \mathbf{A} \theta_t), \quad (45)$$

where

$$\begin{aligned}\mathbf{A} &= \lim_{t \rightarrow \infty} \mathbb{E}[\mathbf{A}_t] \\ &= \lim_{t \rightarrow \infty} \mathbb{E}_\mu [\rho_t \phi_t (\phi_t - \gamma \phi_{t+1})^\top] - \lim_{t \rightarrow \inf ty} \mathbb{E}_\mu [\phi_t] \mathbb{E}_\mu [\rho_t (\phi_t - \gamma \phi_{t+1})^\top]^\top \\ &= \sum_s d_\mu(s) \phi(s) (\phi(s) - \gamma \sum_{s'} [\mathbf{P}_\pi]_{ss'} \phi(s'))^\top - \sum_s d_\mu(s) \phi(s) \\ &\quad * \sum_s d_\mu(s) (\phi(s) - \gamma \sum_{s'} [\mathbf{P}_\pi]_{ss'} \phi(s'))^\top \\ &= \Phi^\top \mathbf{D}_\mu (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi - \Phi^\top \mathbf{d}_\mu \mathbf{d}_\mu^\top (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi \\ &= \Phi^\top (\mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi\end{aligned}\quad (46)$$

6.1. 2-state counterexample

In 2-state counterexample, the matrix of off-policy TD is

$$\begin{aligned}\mathbf{A} &= \Phi^\top \mathbf{D}_\mu (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \times \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \times \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.9 \times \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right) \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= -0.2.\end{aligned}\quad (47)$$

It is negative definite, which is consistent with the divergence of the learning curve of the off-policy TD algorithm.

In 2-state counterexample, the matrix of off-policy CTD is

$$\begin{aligned}\mathbf{A} &= \Phi^\top (\mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \times \left(\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} - \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \right) \\ &\quad \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 0.9 \times \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= 0.25.\end{aligned}\quad (48)$$

It is positive definite, which is consistent with the convergence of the learning curve of the off-policy CTD algorithm.

6.2. Bairds' counterexample

In Bairds' counterexample, the distribution of the states is $\mathbf{D}_\mu = \frac{1}{7} \mathbf{I}$. The transition matrix of the target policy is

$$\mathbf{P}_\pi = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}\quad (49)$$

The feature matrix is

$$\Phi = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (50)$$

Since the dimensionality of the features is 8 and the number of states is 7 – with the former exceeding the latter – we consider the key part of the matrix ignoring the surrounding features.

In Bairds' counterexample, the key matrix of off-policy TD is

$$\mathbf{A}_{\text{key}} = \mathbf{D}_\mu (\mathbf{I} - \gamma \mathbf{P}_\pi) = \frac{1}{7} \mathbf{I} (\mathbf{I} - \gamma \mathbf{P}_\pi) = \frac{1}{7} (\mathbf{I} - \gamma \mathbf{P}_\pi). \quad (51)$$

The eigenvalues $\lambda(\frac{1}{2}(\mathbf{A}_{\text{key}} + \mathbf{A}_{\text{key}}^\top)) \approx (0.2598, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, -0.1155)^\top$. There exist negative eigenvalues for this key matrix, that is, it is not positive definite, which is consistent with the experimental results that the off-policy TD algorithm is divergent.

In Bairds' counterexample, the key matrix of off-policy CTD is

$$\begin{aligned} \mathbf{A}_{\text{key}} &= (\mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi) \\ &= \left(\frac{1}{7} \mathbf{I} - \frac{1}{7} \mathbf{1} \frac{1}{7} \mathbf{1}^\top \right) (\mathbf{I} - \gamma \mathbf{P}_\pi) \\ &= \frac{1}{49} (7\mathbf{I} - \mathbf{1}\mathbf{1}^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi). \end{aligned} \quad (52)$$

The eigenvalues $\lambda(\frac{1}{2}(\mathbf{A}_{\text{key}} + \mathbf{A}_{\text{key}}^\top)) \approx (0.2703, 0.2703, 0.1743, 0.1743, 0.0547, 0.0547, 0)^\top$. The key matrix is positive semi-definite, which is consistent with the fact that Off-policy CTD is convergent in the experimental results.

6.3. General 2-state counterexample

In our attempt to find a counterexample for the non-convergence of the CTD algorithm, we unexpectedly discovered a fact as follows:

Lemma 1. (The matrix of off-policy CTD algorithm in general 2-state example is positive definite) Consider general 2-state example with state features $\Phi = (m, n)^\top$, where $m \neq n$. The behavior policy is given by $\mathbf{P}_\mu = \begin{bmatrix} a & 1-a \\ b & 1-b \end{bmatrix}$, where $0 < a < 1$ and $0 < b < 1$. The target policy is defined as $\mathbf{P}_\pi = \begin{bmatrix} x & 1-x \\ y & 1-y \end{bmatrix}$ where $0 \leq x \leq 1$ and $0 \leq y \leq 1$. In the general example, the matrix of the off-policy CTD algorithm is positive definite.

Proof. The stationary distribution $\mathbf{d}_\mu$ and $\mathbf{P}_\mu$ satisfy the invariance $\mathbf{d}_\mu^\top \mathbf{P}_\mu = \mathbf{d}_\mu^\top$. Then, $\mathbf{d}_\mu = (\frac{b}{1-a+b}, \frac{1-a}{1-a+b})^\top$.

$$\begin{aligned} \mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top &= \frac{(1-a)b}{(1-a+b)^2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \\ (\mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi) &= \frac{(1-a)b}{(1-a+b)^2} (1-x\gamma+y\gamma) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \end{aligned}$$

Then, the matrix of the off-policy CTD algorithm is

$$\mathbf{A} = \Phi^\top (\mathbf{D}_\mu - \mathbf{d}_\mu \mathbf{d}_\mu^\top) (\mathbf{I} - \gamma \mathbf{P}_\pi) \Phi = \frac{(1-a)b}{(1-a+b)^2} (1-x\gamma+y\gamma) (m-n)^2.$$

Since $m \neq n$, the matrix is positive definite. $\square$

7. Conclusion and future work

This paper explores the principles of reward centering, interpreting value-based reward centering as Bellman error centering. We derive the fixed-point solutions under tabular value and linear function approximation. Additionally, we present both on-policy CTD and off-policy CTDC, along with proofs of their convergence.

Future work includes, but is not limited to, (1) extending Bellman error centering to the λ returns; (2) extending Bellman error centering to traditional RL algorithms and their convergence analysis; (3) extending Bellman error centering to policy gradient and actor-critic RL algorithms.

CRediT authorship contribution statement

Xingguo Chen: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis; **Yu Gong:** Investigation, Data curation; **Jinguo Ye:** Validation; **Chao Li:** Funding acquisition; **Shangdong Yang:** Funding acquisition; **Wenhao Wang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Theorem 2

Proof. The proof is based on the multi-time-scale stochastic approximation that is similar to that given by [Sutton et al. \(2009\)](#) for TDC.

For the off-policy CTDC algorithm, the centered one-step linear TD solution is defined as:

$$\begin{aligned} 0 &= \mathbb{E}[(\phi - \gamma \phi' - \mathbb{E}[\phi - \gamma \phi']) \phi^\top] \mathbb{E}[\phi \phi^\top]^{-1} \\ &\quad \mathbb{E}[(\delta - \mathbb{E}[\delta]) \phi] \\ &= \mathbf{A}^\top \mathbf{C}^{-1} (-\mathbf{A} \theta + b). \end{aligned}$$

The matrix $\mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A}$ is positive definite. Thus, the off-policy CTDC algorithm's solution is $\theta_{\text{CTDC}} = \mathbf{A}^{-1} b$.

First, note that recursion (31) and (32) can be rewritten as, respectively,

$$\begin{aligned} \theta_{k+1} &\leftarrow \theta_k + \zeta_k x(k), \\ u_{k+1} &\leftarrow u_k + \beta_k y(k), \end{aligned}$$

where

$$\begin{aligned} x(k) &= \frac{\alpha_k}{\zeta_k} [(\delta_k - \omega_k) \phi_k - \gamma \phi'_k (\phi_k^\top u_k)], \\ y(k) &= \frac{\zeta_k}{\beta_k} [\delta_k - \omega_k - \phi_k^\top u_k] \phi_k. \end{aligned}$$

Recursion (31) can also be rewritten as

$$\theta_{k+1} \leftarrow \theta_k + \beta_k z(k),$$

where

$$z(k) = \frac{\alpha_k}{\beta_k} [(\delta_k - \omega_k) \phi_k - \gamma \phi'_k (\phi_k^\top u_k)].$$

Due to the settings of the step-size schedule $\alpha_k = o(\zeta_k)$, $\zeta_k = o(\beta_k)$, $x(k) \rightarrow 0$, $y(k) \rightarrow 0$, $z(k) \rightarrow 0$ almost surely as $k \rightarrow \infty$. That is the increments in iteration (33) are uniformly larger than those in (32) and the

increments in iteration (32) are uniformly larger than those in (31), thus (33) is the fastest recursion, (32) is the second fast recursion, and (31) is the slower recursion. Along the fastest time scale, iterations of (31)–(33) are associated with the ODEs system as follows:

$$\dot{\theta}(t) = 0, \quad (\text{A.1})$$

$$\dot{u}(t) = 0, \quad (\text{A.2})$$

$$\dot{\omega}(t) = \mathbb{E}[\delta_t | u(t), \theta(t)] - \omega(t). \quad (\text{A.3})$$

Based on the ODE (A.1) and (A.2), both $\theta(t) \equiv \theta$ and $u(t) \equiv u$ when viewed from the fastest timescale. By the Hirsch lemma (Hirsch, 1989), it follows that $\|\theta_k - \theta\| \rightarrow 0$ a.s. as $k \rightarrow \infty$ for some θ that depends on the initial condition θ_0 of recursion (31) and $\|u_k - u\| \rightarrow 0$ a.s. as $k \rightarrow \infty$ for some u that depends on the initial condition u_0 of recursion (32). Thus, the ODE pair (A.1)–(A.3) can be written as

$$\dot{\omega}(t) = \mathbb{E}[\delta_t | u, \theta] - \omega(t). \quad (\text{A.4})$$

Consider the function $h(\omega) = \mathbb{E}[\delta | \theta, u] - \omega$, i.e., the driving vector field of the ODE (A.4). It is easy to find that the function h is Lipschitz with coefficient -1 . Let $h_\infty(\cdot)$ be the function defined by $h_\infty(\omega) = \lim_{r \rightarrow \infty} \frac{h(r\omega)}{r}$. Then $h_\infty(\omega) = -\omega$, is well-defined. For (A.4), $\omega^* = \mathbb{E}[\delta | \theta, u]$ is the unique globally asymptotically stable equilibrium. For the ODE

$$\dot{\omega}(t) = h_\infty(\omega(t)) = -\omega(t), \quad (\text{A.5})$$

apply $\bar{V}(\omega) = (-\omega)^\top(-\omega)/2$ as its associated strict Lyapunov function. Then, the origin of (A.5) is a globally asymptotically stable equilibrium.

Consider now the recursion (33). Let $M_{k+1} = (\delta_k - \omega_k) - \mathbb{E}[(\delta_k - \omega_k) | \mathcal{F}(k)]$, where $\mathcal{F}(k) = \sigma(\omega_l, u_l, \theta_l, l \leq k; \phi_s, \phi'_s, r_s, s < k)$, $k \geq 1$ are the sigma fields generated by $\omega_0, u_0, \theta_0, \omega_{l+1}, u_{l+1}, \theta_{l+1}, \phi_l, \phi'_l, 0 \leq l < k$. It is easy to verify that $M_{k+1}, k \geq 0$ are integrable random variables that satisfy $\mathbb{E}[M_{k+1} | \mathcal{F}(k)] = 0, \forall k \geq 0$. Because ϕ_k, r_k , and ϕ'_k have uniformly bounded second moments, it can be seen that for some constant $c_1 > 0, \forall k \geq 0$,

$$\mathbb{E}[\|M_{k+1}\|^2 | \mathcal{F}(k)] \leq c_1(1 + \|\omega_k\|^2 + \|u_k\|^2 + \|\theta_k\|^2).$$

Now Assumptions (A1) and (A2) of Borkar and Meyn (2000) are verified. Furthermore, Assumptions (TS) of Borkar and Meyn (2000) is satisfied by our conditions on the step-size sequences $\alpha_k, \zeta_k, \beta_k$. Thus, by Theorem 2.2 of Borkar and Meyn (2000) we obtain that $\|\omega_k - \omega^*\| \rightarrow 0$ almost surely as $k \rightarrow \infty$.

Consider now the second time scale recursion (32). Based on the above analysis, (32) can be rewritten as

$$\dot{\theta}(t) = 0, \quad (\text{A.6})$$

$$\dot{u}(t) = \mathbb{E}[(\delta_t - \mathbb{E}[\delta_t | u(t), \theta(t)]) \phi_t | \theta(t)] - \mathbf{C}u(t). \quad (\text{A.7})$$

The ODE (A.6) suggests that $\theta(t) \equiv \theta$ (i.e., a time-invariant parameter) when viewed from the second fast timescale. By the Hirsch lemma (Hirsch, 1989), it follows that $\|\theta_k - \theta\| \rightarrow 0$ a.s. as $k \rightarrow \infty$ for some θ that depends on the initial condition θ_0 of recursion (31).

Consider now the recursion (32). Let $N_{k+1} = ((\delta_k - \mathbb{E}[\delta_k]) - \phi_k \phi_k^\top u_k) - \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k]) - \phi_k \phi_k^\top u_k | \mathcal{I}(k)]$, where $\mathcal{I}(k) = \sigma(u_l, \theta_l, l \leq k; \phi_s, \phi'_s, r_s, s < k)$, $k \geq 1$ are the sigma fields generated by $u_0, \theta_0, u_{l+1}, \theta_{l+1}, \phi_l, \phi'_l, 0 \leq l < k$. It is easy to verify that $N_{k+1}, k \geq 0$ are integrable random variables that satisfy $\mathbb{E}[N_{k+1} | \mathcal{I}(k)] = 0, \forall k \geq 0$. Because ϕ_k, r_k , and ϕ'_k have uniformly bounded second moments, it can be seen that for some constant $c_2 > 0, \forall k \geq 0$,

$$\mathbb{E}[\|N_{k+1}\|^2 | \mathcal{I}(k)] \leq c_2(1 + \|u_k\|^2 + \|\theta_k\|^2).$$

Because $\theta(t) \equiv \theta$ from (A.6), the ODE pair (A.6)–(A.7) can be written as

$$\dot{u}(t) = \mathbb{E}[(\delta_t - \mathbb{E}[\delta_t | \theta]) \phi_t | \theta] - \mathbf{C}u(t). \quad (\text{A.8})$$

Now consider the function $h(u) = \mathbb{E}[\delta_t - \mathbb{E}[\delta_t | \theta] | \theta] - \mathbf{C}u$, i.e., the driving vector field of the ODE (A.8). For (A.8), $u^* = \mathbf{C}^{-1} \mathbb{E}[(\delta - \mathbb{E}[\delta | \theta]) \phi | \theta]$ is the unique globally asymptotically stable equilibrium. Let $h_\infty(u) = -\mathbf{C}u$. For the ODE

$$\dot{u}(t) = h_\infty(u(t)) = -\mathbf{C}u(t), \quad (\text{A.9})$$

the origin of (A.9) is a globally asymptotically stable equilibrium because $\mathbf{C}$ is a positive definite matrix (because it is nonnegative definite and nonsingular). Now Assumptions (A1) and (A2) of Borkar and Meyn (2000) are verified. Furthermore, Assumptions (TS) of Borkar and Meyn (2000) is satisfied by our conditions on the step-size sequences $\alpha_k, \zeta_k, \beta_k$. Thus, by Theorem 2.2 of Borkar and Meyn (2000) we obtain that $\|u_k - u^*\| \rightarrow 0$ almost surely as $k \rightarrow \infty$.

Consider now the slower timescale recursion (31). In the light of the above, (31) can be rewritten as

$$\begin{aligned} \theta_{k+1} &\leftarrow \theta_k + \alpha_k(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k \\ &\quad - \alpha_k \gamma \phi'_k (\phi_k^\top \mathbf{C}^{-1} \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi | \theta_k]). \end{aligned}$$

Let $\mathcal{G}(k) = \sigma(\theta_l, l \leq k; \phi_s, \phi'_s, r_s, s < k)$, $k \geq 1$ be the sigma fields generated by $\theta_0, \theta_{l+1}, \phi_l, \phi'_l, 0 \leq l < k$. Let

$$\begin{aligned} Z_{k+1} &= ((\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k \\ &\quad - \gamma \phi'_k \phi_k^\top \mathbf{C}^{-1} \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi | \theta_k]) \\ &\quad - (\mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k | \theta_k] \\ &\quad - \gamma \phi'_k \phi_k^\top \mathbf{C}^{-1} \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi | \theta_k] | \mathcal{G}(k)) \\ &= ((\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k \\ &\quad - \gamma \phi'_k \phi_k^\top \mathbf{C}^{-1} \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi | \theta_k]) \\ &\quad - (\mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi_k | \theta_k] \\ &\quad - \gamma \mathbb{E}[\phi'_k \phi_k^\top] \mathbf{C}^{-1} \mathbb{E}[(\delta_k - \mathbb{E}[\delta_k | \theta_k]) \phi | \theta_k]). \end{aligned}$$

It is easy to see that $Z_k, k \geq 0$ are integrable random variables and $\mathbb{E}[Z_{k+1} | \mathcal{G}(k)] = 0, \forall k \geq 0$. Further,

$$\mathbb{E}[\|Z_{k+1}\|^2 | \mathcal{G}(k)] \leq c_3(1 + \|\theta_k\|^2), k \geq 0,$$

for some constant $c_3 \geq 0$, again because ϕ_k, r_k , and ϕ'_k have uniformly bounded second moments, it can be seen that for some constant.

Consider now the following ODE associated with (31):

$$\begin{aligned} \dot{\theta}(t) &= \mathbb{E}[(\phi - \gamma \phi' - \mathbb{E}[\phi - \gamma \phi'] \phi^\top) \mathbb{E}[\phi \phi^\top]^{-1} \\ &\quad \mathbb{E}[(\delta - \mathbb{E}[\delta | \theta(t)]) \phi | \theta(t)]. \end{aligned} \quad (\text{A.10})$$

Let

$$\begin{aligned} \bar{h}(\theta(t)) &= \mathbb{E}[(\phi - \gamma \phi' - \mathbb{E}[\phi - \gamma \phi'] \phi^\top) \mathbb{E}[\phi \phi^\top]^{-1} \\ &\quad \mathbb{E}[(\delta - \mathbb{E}[\delta | \theta(t)]) \phi | \theta(t)] \\ &= \mathbf{A}^\top \mathbf{C}^{-1} (-\mathbf{A} \theta(t) + \mathbf{b}), \end{aligned}$$

because $\mathbb{E}[(\delta - \mathbb{E}[\delta | \theta(t)]) \phi | \theta(t)] = -\mathbf{A} \theta(t) + \mathbf{b}$, where $\mathbf{A} = \text{Cov}(\phi, \phi - \gamma \phi')$, $\mathbf{b} = \text{Cov}(r, \phi)$, and $\mathbf{C} = \mathbb{E}[\phi \phi^\top]$.

Therefore, $\theta^* = \mathbf{A}^{-1} \mathbf{b}$ can be seen to be the unique globally asymptotically stable equilibrium for ODE (A.10). Let $\bar{h}_\infty(\theta) = \lim_{r \rightarrow \infty} \frac{\bar{h}(r\theta)}{r}$. Then $\bar{h}_\infty(\theta) = -\mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A} \theta$ is well-defined. Consider now the ODE

$$\dot{\theta}(t) = -\mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A} \theta(t). \quad (\text{A.11})$$

Because $\mathbf{C}^{-1}$ is positive definite and $\mathbf{A}$ has full rank (as it is nonsingular by assumption), the matrix $\mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A}$ is also positive definite. The ODE (A.11) has the origin of its unique globally asymptotically stable equilibrium. Thus, the assumption (A1) and (A2) are verified.

The proof is given above. In the fastest timescale, the parameter w converges to $\mathbb{E}[\delta | u_k, \theta_k]$. In the second fast timescale, the parameter u converges to $\mathbf{C}^{-1} \mathbb{E}[(\delta - \mathbb{E}[\delta | \theta_k]) \phi | \theta_k]$. In the slower timescale, the parameter θ converges to $\mathbf{A}^{-1} \mathbf{b}$. $\square$

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